FUNDAMENTALS OF CALCULUS

Summarized Notes • Concepts, Core Formulas & Essential Graphs

Prepared from the uploaded UIU Math-1151 (Summer 2026) course outline

Primary reference: Calculus: Early Transcendentals, 10th ed., Howard Anton, Irl Bivens & Stephen Davis.
Supplementary reference listed in the syllabus: H. K. Dass, Engineering Mathematics, 15th ed.

A compact study guide for revision. Definitions and formulas are summarized in original wording; exercise/page references in the syllabus are not reproduced as textbook content.

How to use these notes

Follow the syllabus sequence. Each section gives the key idea, formulas, and a basic graph where it helps. Formulas are typeset in a Unicode-capable font; variables use standard mathematical notation. This guide is a summary, not a substitute for worked examples and practice problems.

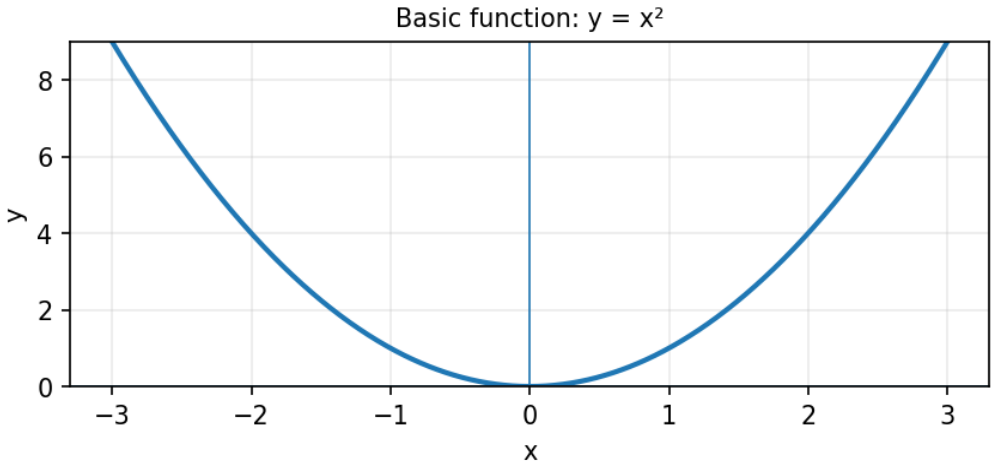
1. Functions, domain and range

Function: a rule assigning each input x in a domain exactly one output y. Write y = f(x). The domain is the allowed set of x-values; the range is the set of resulting y-values. A relation may fail to be a function if one x maps to more than one y.

Domain checks: denominator ≠ 0; for an even root, radicand ≥ 0 (over the reals); for logarithms, argument > 0.

Average rate of change on [a,b]: [f(b) − f(a)] / (b − a), a ≠ b

Example: f(x)=x² has domain ℝ and range [0,∞).



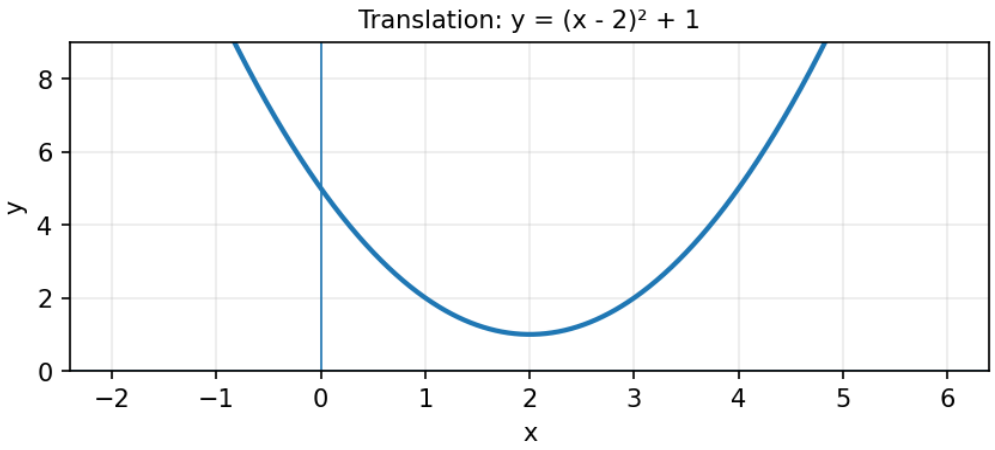
2. Transformations, even and odd functions

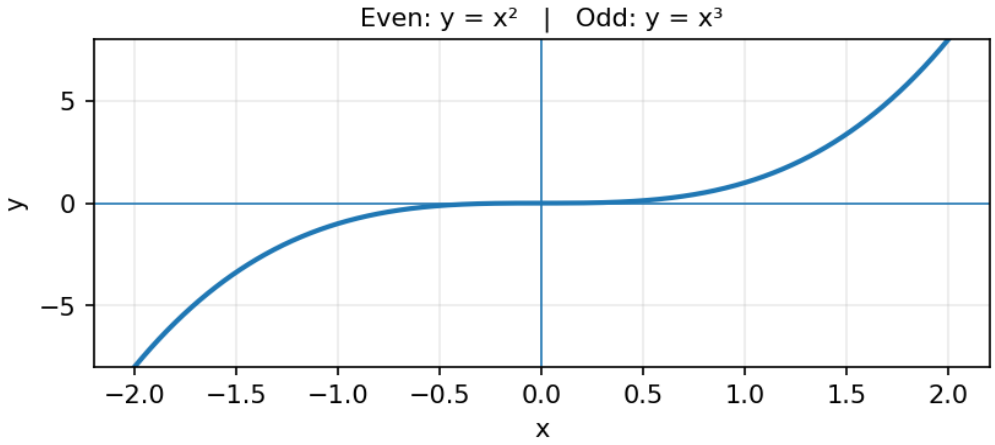
For y=f(x), transformations modify the graph without changing its basic shape.

Vertical shift: y = f(x) + k | Horizontal shift: y = f(x − h)
Vertical scale: y = a f(x) | Horizontal scale: y = f(bx)
Reflection in x-axis: y = −f(x) | Reflection in y-axis: y = f(−x)

For y=f(x−h), shift right h units when h>0. For y=f(x)+k, shift up k units when k>0. In y=f(bx), |b|>1 compresses horizontally; 0<|b|<1 stretches horizontally.

Even: f(−x)=f(x) (y-axis symmetry) | Odd: f(−x)=−f(x) (origin symmetry)





3. Inverse, one-to-one and many-to-one functions

A function is one-to-one if different inputs give different outputs; it passes the horizontal-line test. A many-to-one function has at least two distinct inputs with the same output. An inverse reverses input and output and exists as a function when the original is one-to-one on its domain.

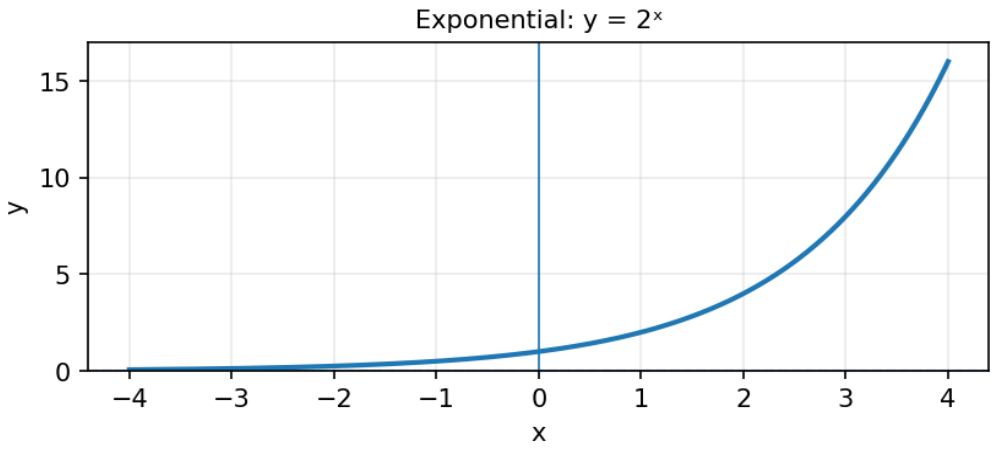
If $y=f(x)$, solve for x in terms of y, then interchange x and y to obtain $f^{-1}(x)$.
$f^{-1}(f(x)) = x$ and $f(f^{-1}(x)) = x$, for inputs in the appropriate domains.

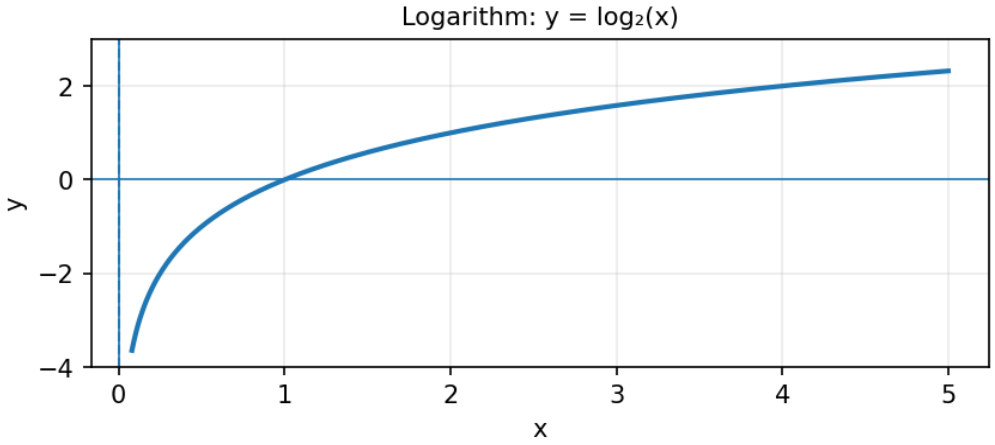
The graphs of f and f^{-1} are reflections across $y=x$. Domain and range swap.

4. Exponential, logarithmic and trigonometric functions

Exponential: $f(x)=a^x$, $a>0$, $a\neq1$. Logarithm: $y=\log_a(x) \Leftrightarrow a^y=x$.
Log laws: $\log_a(xy)=\log_a x+\log_a y$; $\log_a(x/y)=\log_a x-\log_a y$; $\log_a(x^r)=r\log_a x$.
Change of base: $\log_a x = (\ln x)/(\ln a)$. $a^x = e^{x\ln a}$.

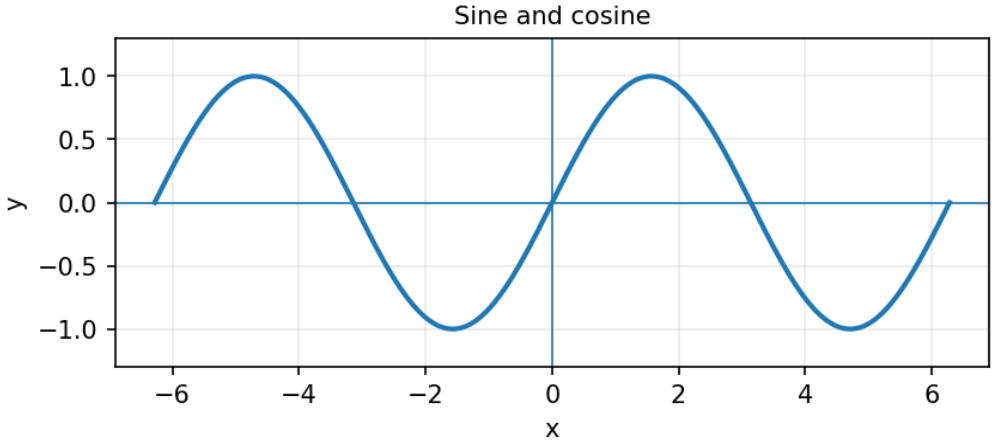
For $a>1$, a^x increases and approaches 0 as $x\to-\infty$; $\log_a x$ is defined only for $x>0$ and has vertical asymptote $x=0$. For $0<a<1$, exponential/logarithmic monotonicity reverses.





$\sin^2 x + \cos^2 x = 1$ | $\tan x = \sin x / \cos x$ (when $\cos x \neq 0$)

$d/dx(\sin x) = \cos x$; $d/dx(\cos x) = -\sin x$ (angles in radians).

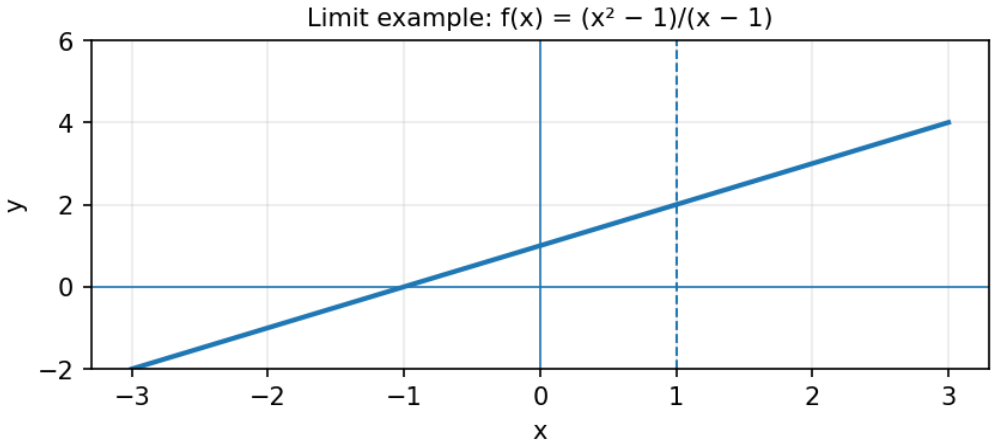


5. Limits and continuity

The limit $\lim_{x\to a} f(x)=L$ means $f(x)$ approaches L as x approaches a; $f(a)$ need not equal L or even be defined. A two-sided limit exists when the left- and right-hand limits exist and are equal.

$\lim_{x\to a}[f(x)\pm g(x)] = L \pm M$; $\lim_{x\to a}[f(x)g(x)] = LM$
$\lim_{x\to a}[f(x)/g(x)] = L/M$, if $M \neq 0$.
Continuity at a: $f(a)$ exists, $\lim_{x\to a} f(x)$ exists, and $\lim_{x\to a} f(x)=f(a)$.

For rational expressions, factor and cancel common factors when evaluating a removable discontinuity (while remembering the original excluded point).



6. Tangent lines, rate of change and differentiation

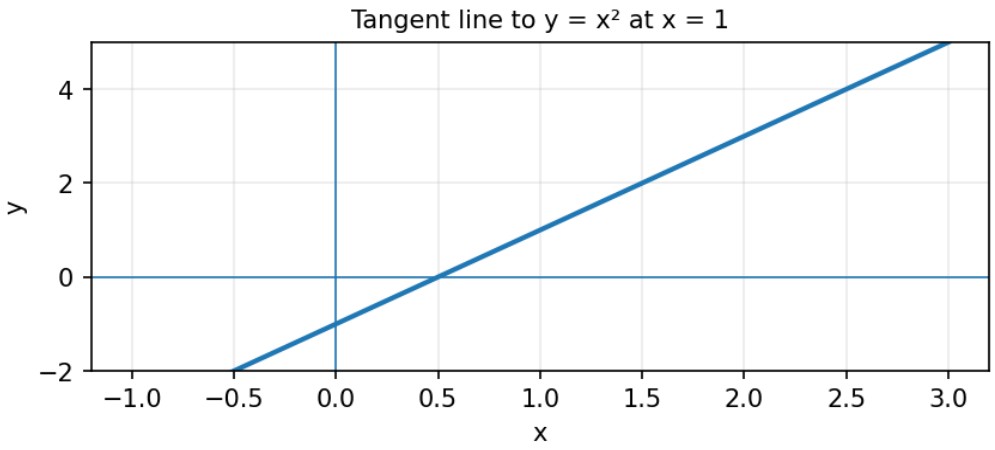
The derivative is the instantaneous rate of change and the slope of the tangent line. The derivative is defined using the limit of secant slopes.

$f'(x) = \lim_{h\to 0}[f(x+h) - f(x)] / h$
Tangent at $x=a$: $y - f(a) = f'(a)(x - a)$
Average velocity $= [s(t_2)-s(t_1)]/(t_2-t_1)$; instantaneous velocity $v(t)=s'(t)$.

Core derivative rules

$d/dx(c)=0$; $d/dx(x^n)=n\,x^{n-1}$; $d/dx[cf]=c\,f'$
$(f\pm g)'=f'\pm g'$; $(fg)'=f'g+fg'$; $(f/g)'=(f'g-fg')/g^2$
Chain rule: $d/dx\, f(g(x)) = f'(g(x))g'(x)$
$d/dx(e^x)=e^x$; $d/dx(a^x)=a^x \ln a$; $d/dx(\ln x)=1/x$; $d/dx(\log_a x)=1/(x \ln a)$
$d/dx(\tan x)=\sec^2 x$; $d/dx(\sec x)=\sec x \tan x$

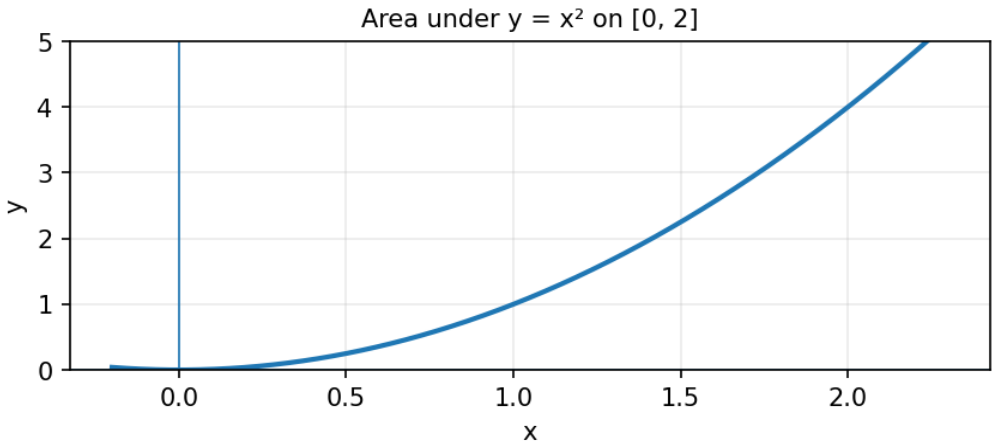
Example: if $y=(3x^2+1)^5$, then $y'=5(3x^2+1)^4\cdot 6x = 30x(3x^2+1)^4$.



7. Area problem and antiderivatives

The area under a nonnegative curve can be approximated by rectangles. Taking finer partitions leads to a definite integral. An antiderivative F of f satisfies $F'(x)=f(x)$.

If $F'(x)=f(x)$, then $\int f(x)\,dx = F(x)+C$.
Power rule ($n\neq -1$): $\int x^n\,dx = x^{n+1}/(n+1)+C$.
$\int 1/x\,dx = \ln\lvert x\rvert+C$; $\int e^x\,dx=e^x+C$; $\int a^x\,dx=a^x/\ln a + C$.



8. Definite integral and Fundamental Theorem of Calculus

A definite integral accumulates signed area over an interval. Geometric area is nonnegative; if a curve lies below the x-axis, its integral contributes negatively.

Fundamental Theorem: if $F' = f$ on $[a,b]$, then $\int_a^b f(x)\,dx = F(b) - F(a)$.

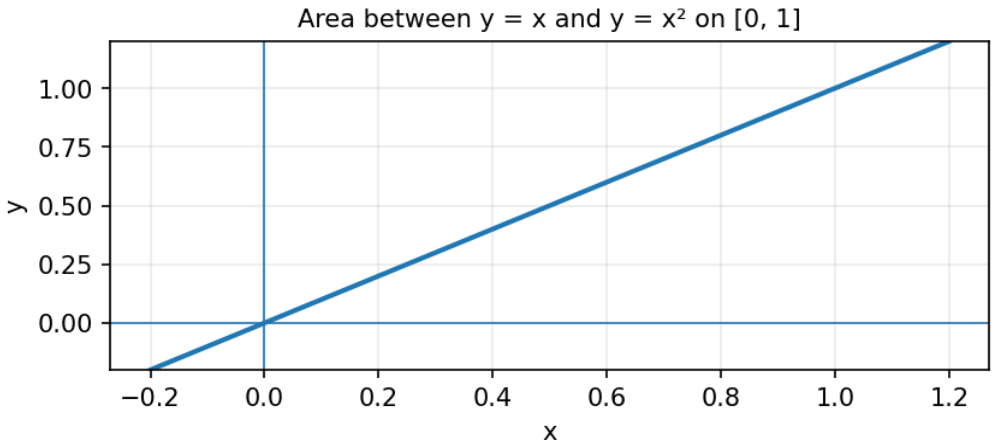
$\int_a^a f(x)\,dx = 0$; $\int_a^b f(x)\,dx = -\int_\beta^a f(x)\,dx$; $\int_a^c f = \int_a^b f + \int_\beta^c f$.

For a continuous f, define $A(x) = \int_a^x f(t)\,dt$. Then $A'(x) = f(x)$.

9. Area between two curves

If upper curve is $y = f(x)$ and lower curve is $y = g(x)$ on $[a,b]$, the area is the integral of upper minus lower. If curves cross, split the interval at intersection points.

$A = \int_a^b [f(x) - g(x)]\,dx$, provided $f(x) \geq g(x)$ on $[a,b]$.



10. Integration methods

Choose a method by the structure of the integrand. Always include +C for indefinite integrals and check by differentiating the result.

Substitution (reverse chain rule)

Let $u = g(x)$, $du = g'(x)dx$; then $\int f(g(x))g'(x)\,dx = \int f(u)\,du$.

Integration by parts

$\int u\,dv = uv - \int v\,du$. (For definite integrals: $\int_a^b u\,v'\,dx = [uv]_a^b - \int_a^b v\,u'\,dx$.)

Trigonometric substitution (common patterns)

$\sqrt{a^2 - x^2}$: $x = a\sin\theta$; $\sqrt{a^2 + x^2}$: $x = a\tan\theta$; $\sqrt{x^2 - a^2}$: $x = a\sec\theta$.

Common integration formulas

$\int \cos x\,dx = \sin x + C$; $\int \sin x\,dx = -\cos x + C$; $\int \sec^2 x\,dx = \tan x + C$.

$\int dx/(a^2 + x^2) = (1/a)\arctan(x/a) + C$ $(a > 0)$.

11. Arc length

For a smooth curve $y = f(x)$, the curve length is approximated by short straight segments; in the limit this yields the arc-length integral.

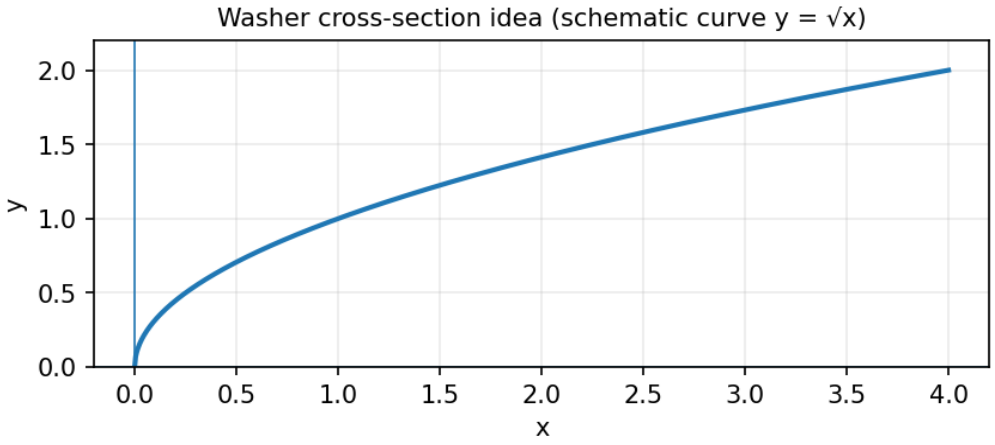
$L = \int_a^b \sqrt{1 + [f'(x)]^2}\,dx$.

For $x=g(y)$, use $L=\int_c^d \sqrt{1+[g'(y)]^2}\,dy$. The derivative and integral must exist appropriately over the interval.

12. Volumes of solids of revolution

When a region is rotated about an axis, cross-sections give disk/washer formulas; cylindrical shells provide another method. Choose the method that makes the radius and integration variable easiest to express.

Disk: $V = \pi\int_a^b [R(x)]^2\,dx$.
Washer: $V = \pi\int_a^b ([R(x)]^2 - [r(x)]^2)\,dx$.
Shells about a vertical axis: $V = 2\pi\int_a^b (\text{radius})(\text{height})\,dx$.



Quick formula sheet

Derivative: $f'(x)=\lim_{h\to 0}[f(x+h)-f(x)]/h$
Power: $(x^n)'=n\,x^{n-1}$ | Chain: $(f\circ g)'=(f'\circ g)g'$
Product: $(fg)'=f'g+fg'$ | Quotient: $(f/g)'=(f'g-fg')/g^2$
Antiderivative: $\int f(x)dx=F(x)+C$ where $F'=f$
FTC: $\int_a^b f(x)dx=F(b)-F(a)$
Area between curves: $\int_a^b(\text{upper}-\text{lower})dx$
Arc length: $\int_a^b\sqrt{(1+[f'(x)]^2)}dx$
Volume: $\pi\int R^2dx$; washer: $\pi\int(R^2-r^2)dx$; shells: $2\pi\int rh\,dx$

Common checks before submitting a solution

- State the domain or restrictions when needed.
- Use radians for trigonometric derivative formulas.
- Keep +C on indefinite integrals.
- For definite integrals, evaluate antiderivative at upper bound minus lower bound.
- For area, identify which curve is upper/lower and split where they cross.
- For inverse functions, check one-to-one behavior or restrict the domain.

Scope and source note

Topic sequence follows the uploaded UIU Math-1151 syllabus: functions and transformations; inverse/exponential/logarithmic functions; limits and continuity; tangent lines and differentiation; area and antiderivatives; definite integrals and area between curves; integration methods; arc length; and volumes of revolution. The uploaded outline explicitly names Anton, Bivens & Davis, Calculus: Early Transcendentals, 10th edition, as the textbook. This document is an independently written summary and does not reproduce textbook pages or worked examples.

Important: This is a compact revision guide. For full rigor and exam readiness, practice the syllabus-listed exercises and confirm any instructor-specific conventions.